

SAJEEVAN (SAJ) VEERIAH

ROBOTICS, MECHATRONICS, AI/ML & END-TO-END AUTOMATION ENGINEER

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PROFILE

Robotics, mechatronics and automation engineer who works across the full stack of a machine: mechanical design, electronics, embedded firmware, controls, autonomy, software and engineering data. Experience spans Ford vehicle software and ADAS validation, GMP process automation and MiR mobile-robot support, automotive emissions and EV testing, IoT field deployment and manufacturing quality. I take systems from requirements and prototyping through integration, test, commissioning and handover, and I am strongest when a fault crosses mechanical, electrical, control, network and software boundaries.

ENGINEERING CAPABILITY

ROBOTICS AND AUTONOMY	ROS 2 Humble, Nav2, MoveIt 2, Gazebo Fortress, SLAM, EKF sensor fusion, localisation, state estimation, motion planning and PID control
EMBEDDED AND ELECTRONICS	ESP32, STM32, FreeRTOS, C/C++, PCB design and bring-up, CAN, UART, I2C, SPI, BLE, sensors, motors, drives and actuators
AUTOMATION AND CONTROLS	Siemens TIA Portal, WinCC, PCS 7, iFIX, PVI+, PLC and HMI/SCADA integration, field devices, FAT, SAT, commissioning and qualification
AI/ML AND ENGINEERING DATA	Python, scikit-learn, OpenCV, YOLO, anomaly detection, predictive-maintenance concepts, time-series analysis, MATLAB, Simulink, InfluxDB and Grafana
MECHANICAL AND VALIDATION	SolidWorks, Fusion 360, engineering drawings and GD&T, prototyping, instrumentation, calibrated testing, fault isolation and design for manufacture
SOFTWARE AND DELIVERY	Python, C/C++, TypeScript, Linux, Git, Docker, CI/CD, REST APIs, PostgreSQL, technical documentation, traceability and handover

ENGINEERING PROJECTS

Autonomous Navigation Rover Independent robotics build, simulation and hardware - Built a differential-drive platform with LiDAR and IMU sensing on ROS 2 Humble and the Nav2 navigation stack. - Implemented SLAM and an extended Kalman filter that fused wheel odometry and IMU data for pose estimation. - Tuned planning, costmaps, controller gains and recovery behaviours, regression-testing changes in Gazebo Fortress before physical deployment. STACK ROS 2 Humble, Nav2, SLAM, EKF, Gazebo Fortress, RViz, Python, C++, Linux	2024 TO 2025
Clinical Ataxia Assessment Device Deakin University, with clinical partners - Designed the ESP32 device, custom PCB and enclosure around four Hall-effect sensors and 100 Hz acquisition. - Implemented real-time recording, Bluetooth app connection, live display and CSV/PDF clinician reporting. - Validated accuracy, direction, reversal, drift and temperature behaviour against reference instruments in MATLAB. STACK ESP32, C/C++, BLE, Altium, MATLAB, embedded sensing, enclosure design	HONOURS CAPSTONE, 2025
ALSO Inventory Scanning Mobile Robot - designed and delivered an operator-support robot now in active end-user use, linking physical stock capture to a controlled inventory-review workflow.	

PROFESSIONAL EXPERIENCE

Automation and Controls Engineer JAG Process Solutions Pty Ltd Process automation and robotics for pharma, biotech and food Brunswick VIC - Delivered GMP control and integration work across PLC logic, HMI/SCADA, field devices, drives and production data through testing, commissioning and handover. - Migrated validated application content from iFIX to PVI+ item by item, checked functional behaviour, and produced FAT, SAT, qualification and handover evidence. - Supported MiR autonomous mobile robot operations and material-flow workflows, tracing faults across robot, controls, network and production interfaces.	JAN 2026 TO JUN 2026
Product Development Test Engineer (Contract) Ford Motor Company via Invenio Global vehicle product development and validation Little River VIC - Validated vehicle software integration and ADAS features across T6 Ranger and Everest using feature-vehicle, breadboard and regression testing for readiness, OTA updates and sign-off evidence. - Instrumented test vehicles and ran structured drives, capturing and analysing CAN data with Vector CANoe and CANalyzer for fault isolation and defect reporting.	OCT 2025 TO JAN 2026
Technical Officer & Quality Assurance ABMARC Automotive emissions, energy, compliance and future mobility South Geelong VIC - Conducted ADR and EURO emissions, EV and PHEV range, and vehicle-systems testing using calibrated instrumentation, data acquisition and CAN tools. - Prepared traceable QA, regulatory, OH&S and test evidence for certification and audit, and maintained emissions-test equipment between programmes.	JUL 2024 TO AUG 2025
Consultant Engineer, IoT and Projects Administrator DuxTel Pty Ltd Networking and low-power IoT telemetry Greater Geelong VIC - Designed and deployed telemetry linking ESP32 devices, sensors, gateways, Linux services and dashboards using LoRaWAN AU915, MQTT, ChirpStack, InfluxDB and Grafana. - Integrated CAN capture, GPS, a custom PCB, condition sensing and MikroTik networking for agricultural equipment monitoring, then ran field trials and handover.	FEB 2024 TO AUG 2024
Production Line Team Lead & Cellar Hand IDL Australia Beverage development, manufacturing and packaging Moorabool VIC Led shift output, changeovers and handover while holding batch quality and traceability, and provided first-level fault finding for line and cellar equipment.	AUG 2022 TO FEB 2024
Production Line Operator and Commissioning Support Carbon Revolution Industry 4.0 carbon-fibre wheel manufacturing Geelong VIC Operated production equipment, recorded in-process quality and traceability data, and supported equipment trials, setup and verification during process changes.	SEP 2021 TO DEC 2021
Undergraduate Quality Engineer Thornton Engineering Australia Structural steel and heavy engineered fabrication Corio VIC Supported drawing review, inspection and test plans, manufacturing data reports, material traceability and AS/NZS compliance across design, workshop and inspection interfaces.	MAR 2018 TO MAR 2020

EDUCATION AND MEMBERSHIP

Bachelor of Mechatronics Engineering (Honours), Distinction. Deakin University, Geelong VIC	2025
Higher National Diploma, Mechatronics, Robotics and Automation Engineering, Distinction. Cardiff Metropolitan University, United Kingdom	2016

MEMBERSHIP, CERTIFICATION & LANGUAGES

Member, Engineers Australia Current Victorian driver license English, Tamil and Sinhala	CURRENT
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